Test Edge Cases and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected edge-test and coverage record.

Probe equality, classify invalid input, run the third-row variation and audit the independent capacity boundary.

Keep one habit: Keep catalogue versions, expected search paths, actual traces and untested conditions visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Test Normal Cases

Design the Decision Process

Humans Set Boundaries

W1 · Isolate the time-equality boundary

> Keep catalogue and quiet requirement fixed for adjacent values.

Your evidence record

- Activity reference: B version 1 in order is B1 Bookmark drawing (10 min, quiet Yes), B2 Story circle (8 min, quiet No), B3 Pattern puzzle (12 min, quiet Yes), B4 Tower challenge (15 min, quiet No). Valid whole minutes 0–30; Quiet required Yes/No. Require time≤minutes AND (quiet not required OR activity quiet), select first fit, no match after all fail; clarify invalid/missing fields before search. P1 resets each run. S0 changes only ≤ to < for time. B version 2 changes only B1 duration to 20. B version 3 uses durations 20,18,17,15 for B1–B4, preserving order and quiet labels. Keep copies separately labelled.

- For catalogue B version 1, write expected **output**s and reasons for 9 Yes, 10 Yes and 11 Yes. Explain why keeping quiet and catalogue fixed helps examine the time **boundary**.

  Predict what the faulty strictly-less-than S comparison does for 10 Yes.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Check validation before search

> Distinguish accepted zero from blank, fractional and invalid inputs.

Your evidence record

- Classify these requests before searching: 0 Yes, 30 Yes, 31 Yes, blank minutes with Yes, 10.5 Yes, and 10 Maybe. Give each expected **output** or clarification and explain whether any candidate should be checked.

  Why must zero not be treated like a blank?

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Run the labelled third-row variation

> Use B version 2 consistently for both methods and expectations.

Your evidence record

- Copy B version 1 and change only B1's duration to 20, labelling it B version 2. Write the expected result for 12 Yes, then run your saved P1 and the manual baseline on this copy. Record the actual traces separately.

  Explain why an answer checked against B version 1 would not be a fair judgement of this run.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Audit the catalogue-version label

> Use the supplied B version 3 reference and preserve what actually ran.

Your evidence record

- For W5, B version 3 has durations B1=20, B2=18, B3=17, B4=15 minutes, with original quiet labels Yes,No,Yes,No and original order. B version 1 remains 10,8,12,15. Both use the same input and first-fit contract.

- A team runs 15 No on B version 3, gets B4, but records “B version 1” and calls B4 correct. Explain the documentation error and the expected result under each catalogue.

  State what must stay the same in a fair prototype-versus-baseline comparison and why a test not yet run cannot be counted as a pass.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Find an independent capacity boundary

> Keep one first attempt before hints or feedback.

Read and explain independently

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Find an independent capacity boundary

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Expand the sorter test suite with edge, missing-data, and fairness cases plus expected safe behavior.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Check the validation **boundary** for Testing boundaries and varied conditions.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Testing boundaries and varied conditions. Proposed explanation: “At the limit means outside the rule”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A sorter handles clean paper but fails on crumpled paper and mixed packaging.

Design revealing tests.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorter works on clear photos. Test a blurred pencil, a dim book and an unfamiliar lunch box. What makes these useful tests?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 34 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Before testing, define outcomes for clean paper, metal foil, mixed packaging and an unreadable card. The team later changes mixed packaging's expected result to match its sorter. Explain the evaluation problem.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A new equipment selector lists E1 first, capacity 4; E2 second, capacity 8. Valid group size is a whole number from 1 to 10.

Choose the first item whose capacity is at least the group size, or no match; **invalid input** needs clarification. A faulty version uses capacity strictly greater than group size.

Independently give correct and faulty **output**s for groups 3, 4 and 5, then for group 8. Explain the defect and give the correct response to group 0. No booking occurs.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

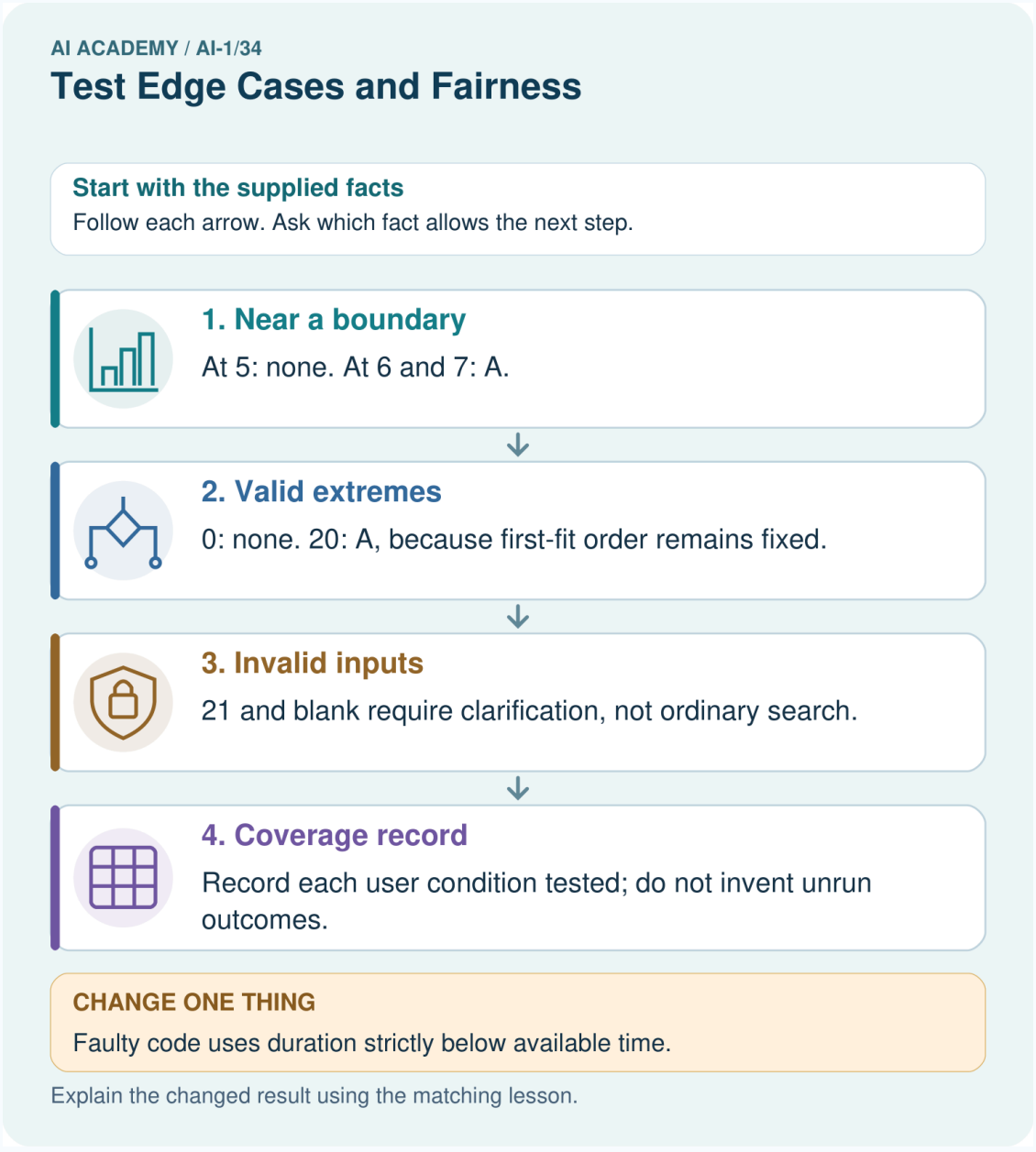
> Keep the explanation and the support you used.

Use this when helpful

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which single **boundary** run most directly detects the strict-comparison bug, and why?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.
- Which labelled step matches this detail? Record each user condition tested; do not invent unrun outcomes.
- 1. Near a **boundary**
- 2. **Invalid input**s

- 3. Coverage record
- 4. Valid extremes
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Coverage record Record each user condition tested; do not invent unrun outcomes. Adjacent values below, at and above 6 expose an accidental strict inequality.

Zero is valid even though no activity fits. The top valid **input** does not require selecting the longest activity: catalogue order still controls selection.

Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Changed-case question

- Changed-case practice: Faulty code uses duration strictly below available time. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Request 6 now wrongly returns none.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Request 6 now wrongly returns none. A's duration equals the available time and should be eligible under at most.

Repair and save your work

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Test Edge Cases and Fairness: the case

AI ACADEMY / AI-1/34 · 01 · LOOK AT THE FACTS

Test Edge Cases and Fairness: the case

The situation

A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20.

Supplied evidence

Booklet A takes 6 minutes; B takes 12. Return first duration at most available time. Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

By the end, I can design boundary and coverage tests instead of assuming routine success is enough.

Follow the labelled picture with the matching explanation. Open the vector diagram

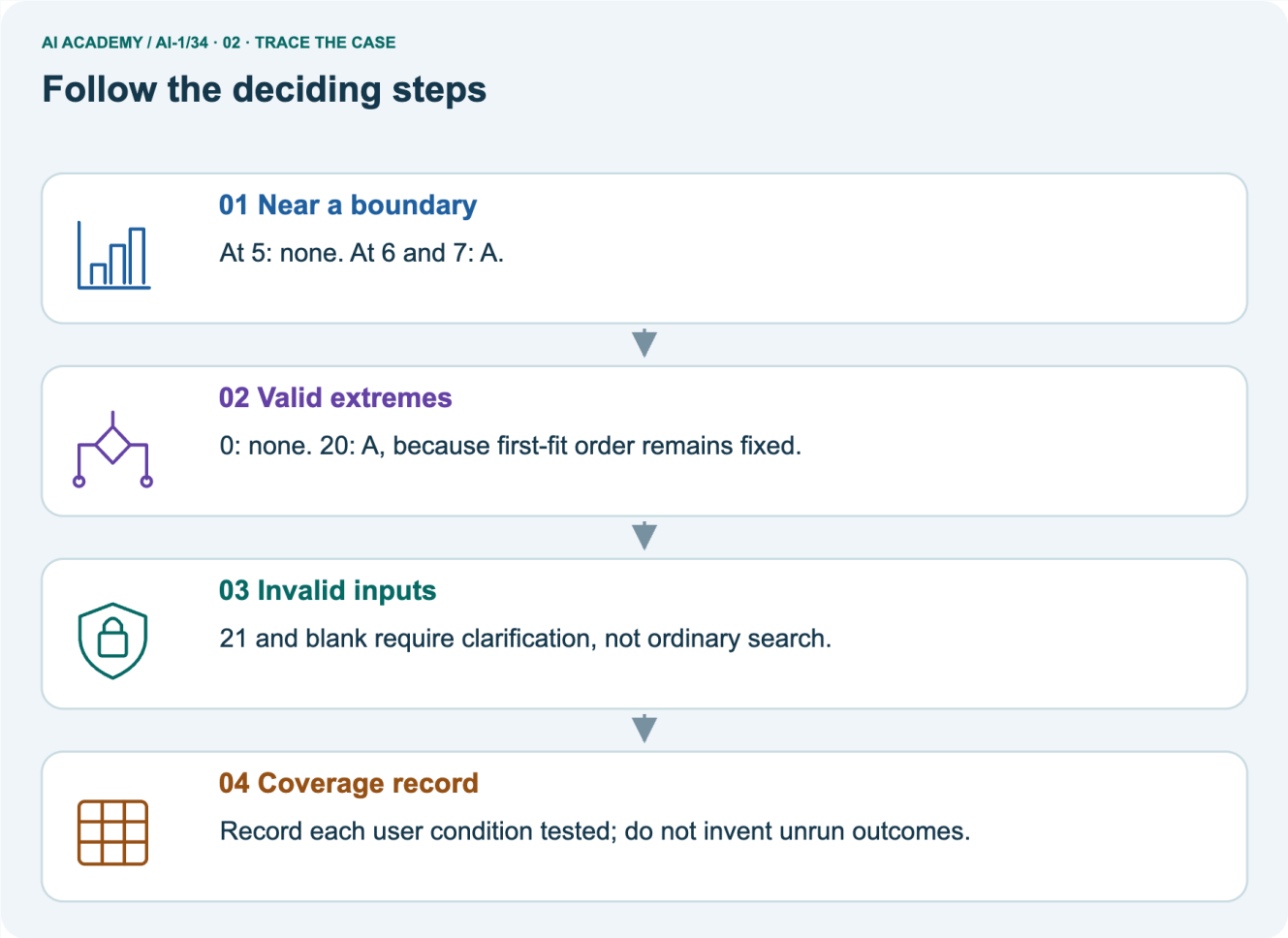
A fictional craft-book chooser accepts requests for whole available minutes from 0 to 20. Supplied facts: Booklet A takes 6 minutes; B takes 12. Return first duration at most available time.

Check 5,6,7; also 0,20,21 and blank. Same catalogue applies to every user.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Near a **boundary**: At 5: none.

At 6 and 7: A. → Valid extremes: 0: none. 20: A, because first-fit order remains fixed. → **Invalid input**s: 21 and blank require clarification, not ordinary search. → Coverage record: Record each user condition tested; do not invent unrun outcomes.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three requests around ten minutes

AI ACADEMY / AI-1/34 · 03 · INSPECT THE LESSON MODEL

Three requests around ten minutes

Available time	Check B1 time	Result for quiet Yes
9 minutes	10 is too long	No match in catalogue B
10 minutes	10 fits exactly	Suggest B1
11 minutes	10 fits with time left	Suggest B1

Keep Quiet required Yes and catalogue B version 1 fixed while changing available time.

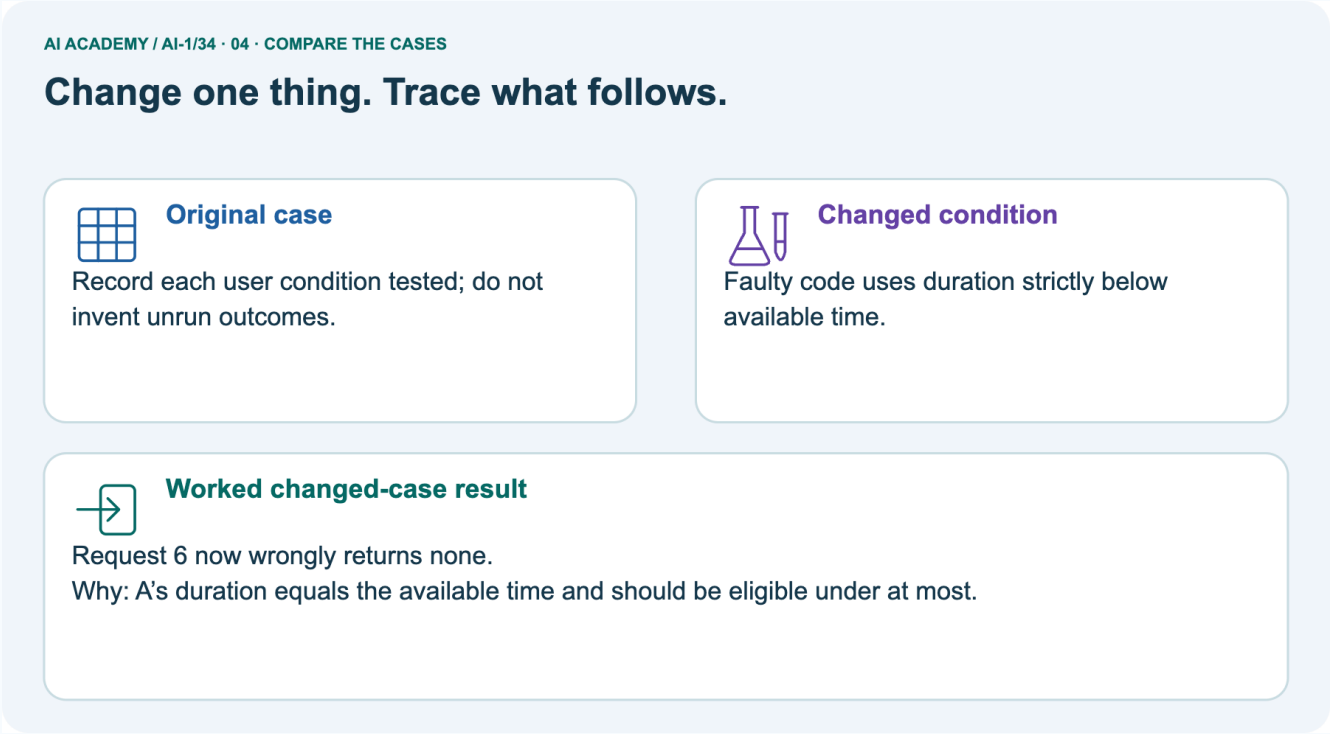
Follow the labelled picture with the matching explanation. Open the vector diagram

Three requests around ten minutes. 9 minutes | 10 is too long | No match in catalogue B. 10 minutes | 10 fits exactly | Suggest B1. 11 minutes | 10 fits with time left | Suggest B1

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

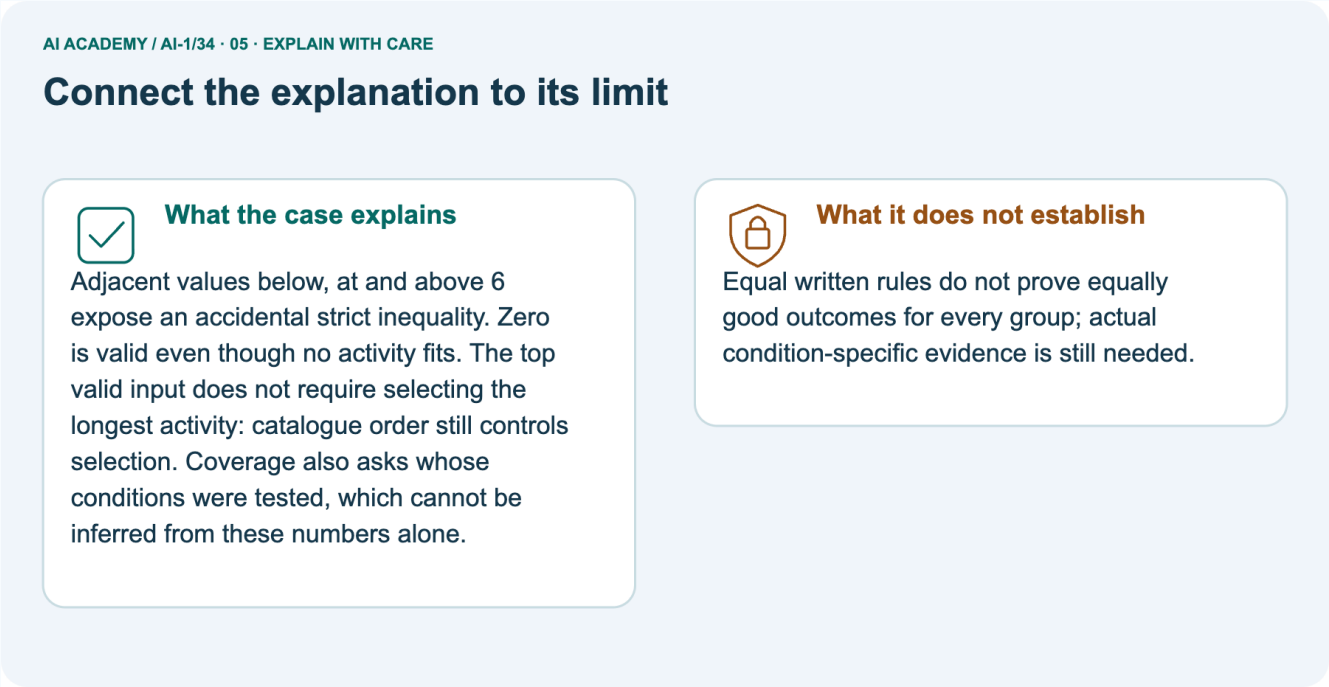
Original case: Record each user condition tested; do not invent unrun outcomes. Changed condition: Faulty code uses duration strictly below available time. Expected result: Request 6 now wrongly returns none.

Reason: A's duration equals the available time and should be eligible under at most.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adjacent values below, at and above 6 expose an accidental strict inequality. Zero is valid even though no activity fits.

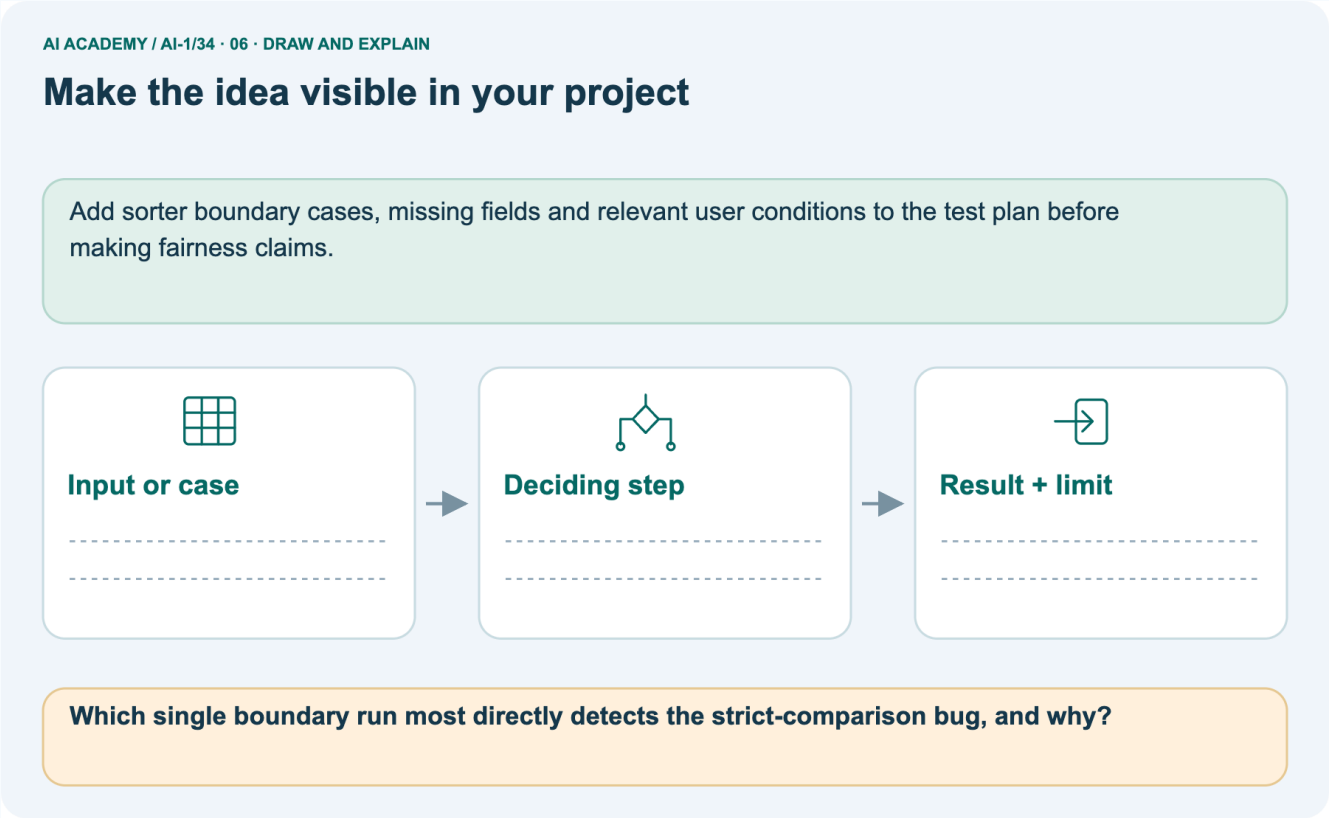
The top valid **input** does not require selecting the longest activity: catalogue order still controls selection. Coverage also asks whose conditions were tested, which cannot be inferred from these numbers alone.

Limit: Equal **written rule**s do not prove equally good outcomes for every group; actual condition-specific **evidence** is still needed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add sorter **boundary** cases, missing fields and relevant user conditions to the test plan before making fairness claims. Draw your **input**, deciding step, result and limit.

Supported question: Which single **boundary** run most directly detects the strict-comparison bug, and why?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Test Edge Cases and Fairness: the case

02. Follow the deciding steps

03. Three requests around ten minutes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.